1 Sensitivity Analysis of Vehicle-Average and Order-Specific Predictors

The pickup-time and detour-ratio predictors adopted in the main framework use vehicle-pool average targets. This design is consistent with the route-aggregated, pre-routing structure of A-CAM, because the realized order-specific service attributes are unavailable before a pickup–delivery sequence is selected. Nevertheless, assigning the same predicted average to all orders within a candidate pool may omit part of the within-pool heterogeneity. This supplementary appendix therefore examines whether direct order-specific predictors improve prediction and downstream optimization performance.

1.1 Order-Specific Predictor Construction and Predictive Performance

We estimate direct order-specific linear predictors $\widehat{w}_{rk}$ and $\widehat{e}_{rk}$ for each order r and vehicle k. These predictors use only information available before route sequencing, including the vehicle-to-pickup distance, the order's direct-trip distance and direction, its relative geometric position within the candidate pool, pairwise pickup and drop-off distances, and the original pool-level features. They do not use the realized pickup–delivery sequence or realized route times. Accordingly, they provide an implementable order-specific sensitivity specification rather than fully route-aware predictors. Therefore, this experiment should be interpreted as a sensitivity analysis of predictor granularity and downstream executable objective quality, rather than as an exact optimality-gap calculation relative to a fully route-aware prediction model.

Both predictor specifications are evaluated against the same order-level realized targets. For the vehicle-average specification, the predicted vehicle-pool average is assigned to each order in the corresponding pool. This comparison differs from Table 7 of the main paper, whose dependent variables are vehicle-pool averages. Table 1 reports the order-level results for the nonnegative predictor versions used in the optimization experiments.

Table 1: Order-level predictive performance of vehicle-average and order-specific linear predictors

Attribute	Predictor	MAE	RMSE
Pickup time	Vehicle-average prediction assigned to each order	9.007	12.482
Pickup time	Direct order-specific linear prediction	8.103	11.081
Detour ratio	Vehicle-average prediction assigned to each order	0.083	0.132
Detour ratio	Direct order-specific linear prediction	0.059	0.096

Relative to assigning the vehicle-average prediction to each order, the order-specific predictor reduces pickup-time MAE and RMSE by 10.04% and 11.22%, respectively, and reduces detour-ratio MAE and RMSE by 29.08% and 27.56%, respectively. These improvements confirm that the pre-routing order-level features capture part of the within-pool heterogeneity. The remaining prediction errors reflect, among other factors, the dependence of the realized service attributes on the subsequently selected pickup–delivery sequence.

1.2 Sensitivity of A-CAM+PO

We next embed the vehicle-average and order-specific predictors into A-CAM. The experiment uses ten fixed cases from each of A_{4_2}, A_{6_3}, A_{8_4}, A_{10_5}, and A_{15_8}. The cases are fixed independently

of the comparison outcomes. Both formulations use the same data, algorithmic settings, and 60-second solver limit. Their returned matching decisions are evaluated using the same Policy 2 route-and-pricing post-optimization procedure. Accordingly, the reported objectives are postprocessed executable objective values rather than the internal approximate master-model objectives.

For each paired case, define

$$\Delta\text{Obj} = 100\,\frac{\text{Obj}^O - \text{Obj}^V}{\text{Obj}^V}, \qquad \Delta\text{Time} = 100\,\frac{\text{Time}^O - \text{Time}^V}{\text{Time}^V},$$

where superscripts V and O denote the vehicle-average and order-specific predictors, respectively. Table 2 reports the means of these case-level paired changes.

Table 2: A-CAM+PO sensitivity to vehicle-average and order-specific predictors

	Mean objective		Mean paired	Mean time (s)		Mean paired	Variable	Linear-constraint
Instance group	V-based	O-based	ΔObj (%)	V-based	O-based	ΔTime (%)	increase (%)	increase (%)
A_{4_2}	49.85	50.22	0.75	0.78	1.00	34.11	23.36	20.78
A_{6_3}	76.22	76.75	0.65	3.34	5.01	58.49	26.57	22.93
A_{8_4}	100.48	101.87	1.38	53.81	60.00	–	28.51	24.15
A_{10_5}	122.15	124.88	2.30	60.00	60.00	–	29.81	24.94
A_{15_8}	184.80	183.53	-0.68	60.00	60.00	–	31.81	26.09
Mean	–	–	**0.88**	–	–	**46.30**	**30.03**	**25.05**

Notes: Each instance group contains ten cases. V-based and O-based denote the vehicle-average-based and order-specific-based predictors, respectively. The paired time change is omitted for A_{8_4} and the larger groups because at least one formulation reaches the 60-second time limit. The mean ΔTime is therefore calculated from the 20 uncensored A_{4_2} and A_{6_3} cases.

Across the 50 paired cases, the order-specific A-CAM formulation improves the postprocessed objective by 0.88% on average. It produces a higher objective in 23 cases, the same objective in 12 cases, and a lower objective in 15 cases. For A_{4_2} and A_{6_3}, all cases under both predictor specifications finish within the 60-second limit. At A_{8_4}, seven vehicle-average cases finish within the limit, whereas none of the order-specific cases do. For A_{10_5} and A_{15_8}, neither formulation finishes within the limit in any case. For the 20 uncensored A_{4_2} and A_{6_3} cases, the mean paired runtime increase is 46.30%. The order-specific formulation also increases the number of variables by 23.36%–31.81% and the number of linear constraints by 20.78%–26.09%.

The computational effect becomes more pronounced as the instance size increases. At A_{10_5} and A_{15_8}, both formulations reach the 60-second limit, so their identical recorded times are censored observations rather than evidence of equal computational difficulty. Under the fixed real-time budget, the order-specific objective improvement is positive for A_{10_5} but reverses for A_{15_8}, where the mean postprocessed objective is 0.68% lower. These non-monotone changes are consistent with the discrete nature of matching decisions: a more accurate local prediction does not guarantee a uniformly better matching within a limited search time.

1.3 Sensitivity of PAD^2

We further embed the two predictor specifications into PAD^2. The order-specific predictions are evaluated procedurally for each vehicle-pool/column considered by the algorithm and are then used in the acceptance calculation. This implementation incurs additional predictor evaluations but does not introduce the same collection of vehicle–order prediction variables and constraints into a single monolithic formulation.

Table 3: PAD^2 sensitivity to vehicle-average and order-specific predictors

	Mean objective		Mean paired	Mean time (s)		Mean paired	Order-specific outcome		
Instance group	V-based	O-based	ΔObj (%)	V-based	O-based	ΔTime (%)	Higher	Equal	Lower
A_{4_2}	50.82	51.18	0.91	0.22	0.23	2.41	3	5	2
A_{6_3}	75.98	77.70	2.20	1.03	1.12	9.75	5	5	0
A_{8_4}	100.24	103.64	3.51	3.32	3.35	2.30	7	1	2
A_{10_5}	126.36	127.33	0.90	9.83	9.88	2.73	7	0	3
A_{15_8}	188.91	198.18	4.92	23.40	24.26	6.30	8	0	2
Mean	–	–	**2.49**	–	–	**4.70**	–	–	–

Notes: Each instance group contains ten cases. V-based and O-based denote the vehicle-average-based and order-specific-based predictors, respectively.

For PAD^2, the order-specific predictors increase the objective by 2.49% on average and increase the paired runtime by 4.70% on average. Across the 50 paired cases, the order-specific specification produces a higher objective in 30 cases, the same objective in 11 cases, and a lower objective in 9 cases. The group-level runtime increase ranges from 2.30% to 9.75%.

Taken together, the results support retaining the vehicle-average predictors in the main A-CAM formulation as a deliberate accuracy–tractability choice. The order-specific sensitivity analysis confirms that vehicle averages omit part of the within-pool heterogeneity, but their replacement yields only a modest and non-uniform improvement in A-CAM solution quality while increasing the formulation size and computational burden. At the same time, PAD^2 can incorporate the tested order-specific predictors with a relatively small average runtime increase. Developing richer pre-routing or route-aware predictors within PAD^2 remains a promising direction for future research.